Newumal Weerasinghe

Team Lead & Associate Technical Lead | Full-Stack Engineering | Production AI

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PROFESSIONAL SUMMARY

Hands-on engineering leader with 7+ years building and scaling production SaaS across React.js, Node.js, AWS, PostgreSQL and NoSQL. Combines full-stack engineering with team leadership, system design, SaaS integrations, and performance and cloud cost optimization. Ships reliable, cost-efficient production AI/LLM capabilities, from RAG and context graphs to custom, tool-using agents and Copilot workflows, while keeping human ownership of engineering decisions.

WORK EXPERIENCE

Associate Technical Lead & Team Lead

Jan 2026 – Present

Velaris | Moratuwa, Sri Lanka

- Lead a cross-functional engineering team of 5 engineers, owning sprint planning, technical direction, code reviews, mentoring, release delivery and engineering interviews, and contributing to system design.
- Designed, planned and built a cost-efficient, scalable onboarding AI agent that fast-tracks client onboarding by reducing manual setup work and cutting the time users spend getting set up, using Pydantic AI, MCP-based tools, LLM routing and multi-step agent workflows.
- Turned the onboarding agent into a product-expansion opportunity, opening an upsell and revenue-growth angle by making faster, lower-touch onboarding a differentiator for the platform.
- Proposed and led an AI feature from proof-of-concept to production, securing backing from the CEO, Head of AI and product owners; partnered with the AI team to generalize the approach, which was later reused across other teams for sales and product workflows.
- Introduced a data-driven approach to churn and reliability work, documenting root causes, durable fixes, and the engineering and QA effort involved, which aligned Product and Engineering on priorities, freed capacity for new features, and was later adopted by other teams.
- Led stability, scalability and cost initiatives across the customer 360-degree view and a major (v4) platform rollout, scaling workloads to millions of records, improving critical-path performance by 60% and reducing AWS operational cost by 20%.
- Mentored engineers across multiple teams on email, calendar and transcript integrations, testing strategy and production delivery, and supported hiring through technical interviews.
- Established responsible AI engineering practices through hands-on sessions, standards and custom agents designed for accurate, low-hallucination output, driving adoption across the engineering team.

Senior Software Engineer (Full Stack)

Jul 2023 – Jan 2026

Velaris | Moratuwa, Sri Lanka

- Built an AI Copilot capability that turns natural-language questions into line, pie, bar and KPI charts plus interactive Canvas dashboards, and extended Copilot context to handle more detailed user questions.
- Built user-facing AI features across multiple LLMs using RAG, context graphs and agentic systems, with a focus on production reliability and reducing LLM cost by 30%.
- Reworked core email processing and event filtering, improving synchronization and processing performance with no bugs reported after release.
- Led AWS cost-optimization work across compute, I/O, database usage and logging, and built token-health monitoring to detect credential revocations and alert before customer impact.
- Served as acting team lead during a senior lead's absence, running Scrum ceremonies, distributing work, coordinating with QA and Product, and shipping multiple stable production releases.

Software Engineer (Full Stack)

Jul 2022 – Jun 2023

Velaris | Moratuwa, Sri Lanka

- Led user-integration projects end to end across Google Cloud and Microsoft Azure, including OAuth app registration, consent screens, verification reviews, token/authentication flows, Gmail, Outlook, calendars and meeting transcripts.
- Designed and shipped microservice pipelines and full-stack features for Microsoft Teams transcripts, Google and Outlook calendars and Gmail, delivering the Teams integration on a tight deadline with no reported post-release issues.
- Built and delivered a Salesforce integration suite of five components against three originally planned, including streaming-data handling for large datasets with very few defects.

Software Engineer (Full Stack) Mar 2021 – Jul 2022

Sezenta (Pvt.) Ltd | Battaramulla, Sri Lanka

- Built web and cross-platform mobile applications using React.js, React Native, Node.js, MongoDB and AWS, including REST APIs with Express.js and NestJS.
- Improved application scalability and performance through AWS services and backend design changes.

Development Engineer & Co-Founder Feb 2019 – Mar 2021

STEM Labs (Pvt.) Ltd | Nugegoda, Sri Lanka

- Delivered IT and STEM education programmes covering robotics and electronics, while leading web design and development projects using PHP, JavaScript, CSS and WordPress.
- Developed software and hardware solutions, including an IoT hardware product for which a patent application was filed.

Associate Software Engineer Feb 2017 – Feb 2019

Effective Solutions (Pvt.) Ltd | Moratuwa, Sri Lanka

- Built Symfony/PHP software including a loyalty platform and a student enrollment system for Trinity College, and developed e-commerce solutions for clients including Softlogic and TyreOnline.
- Designed IoT solutions using Arduino, MQTT, Node.js and Node-RED, including GPS-tracking integrations via REST APIs.

TECHNICAL SKILLS

Leadership & Architecture: Technical leadership, Team management, System design, Microservices, Caching, Queues / background processing, Performance optimization, Scalability, Code review, Mentoring, Agile/Scrum, Technical interviews

Backend & Frontend: JavaScript (ES6+), TypeScript, Node.js, Express.js, NestJS, React.js, Next.js, Redux, React Native, SSR/SSG (basic), PHP, Python

Cloud & DevOps: AWS (Lambda, EC2, S3, SQS), AWS (Step Functions, CloudWatch, CloudFormation), Docker, CI/CD, Google Cloud Platform, Microsoft Azure

Data, Identity & Integrations: PostgreSQL, MongoDB, DynamoDB, MySQL, TypeORM, REST APIs, OAuth 2.0, JWT, SSO, API security, Authorization / RBAC, Google / Microsoft / Salesforce integrations

AI Engineering: LLM integration, RAG, Context graphs, Custom AI agents, Prompt engineering, Evals, LLM cost management, AI-assisted SDLC (GitHub Copilot, Cursor, Claude Code), Pydantic AI, MCP (Model Context Protocol), Tool calling, LLM routing

Testing & Tools: Automated testing, Jest, ESLint, Git, JIRA

EDUCATION

BSc in Electronics & IT

University of Colombo | Colombo, Sri Lanka

ACHIEVEMENTS

Technical Content Creator: Publishes engineering and technical-leadership content on YouTube (@buildwithnewumal).

Kotiyak Watina Adahasak, Season 2 (2018): Reached the final 24 of a national innovative-ideas competition broadcast on Sirasa TV, Sri Lanka.